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**PROJECT PROPOSAL**

**TITLE:** DESIGN AND IMPLEMENTATION OF AN INTEGRATED WEB-BASED AGRICULTURAL ECOSYSTEM: A SOLUTION FOR MARKET ACCESS, AUTOMATED DISEASE DIAGNOSTICS AND AGRONOMIC ADVISORY

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## Introduction

The agricultural sector is widely recognized as a catalyst for economic growth and poverty reduction in developing nations. However, the sector is currently undergoing a "digital divide" where the benefits of technology are not reaching the farmers who need them most. Addison et al. (2024) highlight that while Ghana is championing digital transformation, rural farmers are often left behind due to a lack of tailored tools that address their specific livelihood strategies.

Traditionally, farmers operate in information silos. A crop farmer may struggle with determining the precise fertilizer type for their soil, while a poultry farmer may lose an entire flock to disease because veterinary officers are scarce in rural districts. Furthermore, even when production is successful, the market mechanism often fails the farmer. Seerangan and Venkata Ravi (2024) argue that the lack of a structured, transparent marketplace forces farmers to sell produce at throwaway prices to middlemen who control the supply chain.

Recent advancements in Artificial Intelligence (AI) and Web Technologies offer a pathway to bridge these gaps. AI-driven precision agriculture has become a game-changing technology, allowing for early disease identification and yield optimization through computer vision and predictive modelling (Babatunde et al., 2025). This project seeks to harness these technologies to create a holistic "Smart-Farm" ecosystem.

## Problem Statement

Despite the availability of mobile phones in rural areas, Ghanaian farmers face three interconnected problems that existing solutions fail to address comprehensively:

1. **Economic Isolation (Market Access):** Smallholder farmers lack a centralized platform to showcase their produce. They rely on physical markets and middlemen, leading to price exploitation. Seerangan and Venkata Ravi (2024) emphasize that E-Agri commerce is essential to bypass these traditional barriers and ensure fair pricing.
2. **Diagnostic Latency (Veterinary Gap):** Poultry farming is highly vulnerable to infectious diseases like Coccidiosis. Traditional detection relies on manual observation or distant labs, which is slow and error-prone. Sambo et al. (2025) note that the recent progress in machine learning offers a promising alternative for early detection, yet few accessible tools exist for the average farmer.
3. **Information Asymmetry (Advisory Gap):** Extension officers cannot visit every farm. Farmers often lack immediate answers to daily questions regarding fertilizer application or crop suitability. Khaliq et al. (2025) identify that traditional farming methods dependent on generalized practices often lead to inefficient resource utilization and reduced soil health.

## Research Questions & Objectives

### 5.1 General Objective

To design and implement an integrated web-based platform that facilitates agricultural e-commerce and provides AI-driven diagnostic and advisory support to crop and poultry farmers.

### 5.2 Specific Objectives

1. **To develop an e-commerce module** that allows farmers to list produce and connect directly with buyers, securing better market prices.
2. **To implement a Machine Learning model** (using Convolutional Neural Networks) that detects Coccidiosis by analyzing images of poultry droppings.
3. **To create an interactive Chatbot** that provides agronomic advice (fertilizers, crops) and poultry management tips to users.
4. **To evaluate the system** for usability and accuracy, ensuring it meets the needs of non-technical rural users.

## Significance of the Study

This study is significant because it moves beyond theoretical research into practical application.

- **For the Farmer:** It directly impacts livelihoods. Addison et al. (2024) found that digital adoption correlates with improved financial, human, and social capital for farmers in Ghana.
- **For the Agriculture Sector:** It democratizes access to veterinary expertise. By implementing lightweight models as suggested by Islam et al. (2025), this tool brings high-end diagnostics to low-resource environments.
- **For Academia:** It contributes to the field of "Precision Agriculture," demonstrating how integrated approaches (soil, disease, and market) can function in a unified system, a need highlighted by Khaliq et al. (2025).

## Literature Review

**7.1 E-Commerce in Agriculture** Digitalization has transformed agricultural markets. Seerangan and Venkata Ravi (2024) explain that "E-Agri Commerce" transforms the market by enhancing transparency and efficiency. Their research confirms that digital platforms allow farmers to bypass intermediaries, directly impacting their profit margins.

**7.2 AI in Poultry Disease Detection** Poultry diseases like Coccidiosis cause significant economic damage. Sambo et al. (2025) reviewed various ML approaches and concluded that

Deep Learning, specifically Convolutional Neural Networks (CNNs), significantly outperforms traditional methods in disease identification.

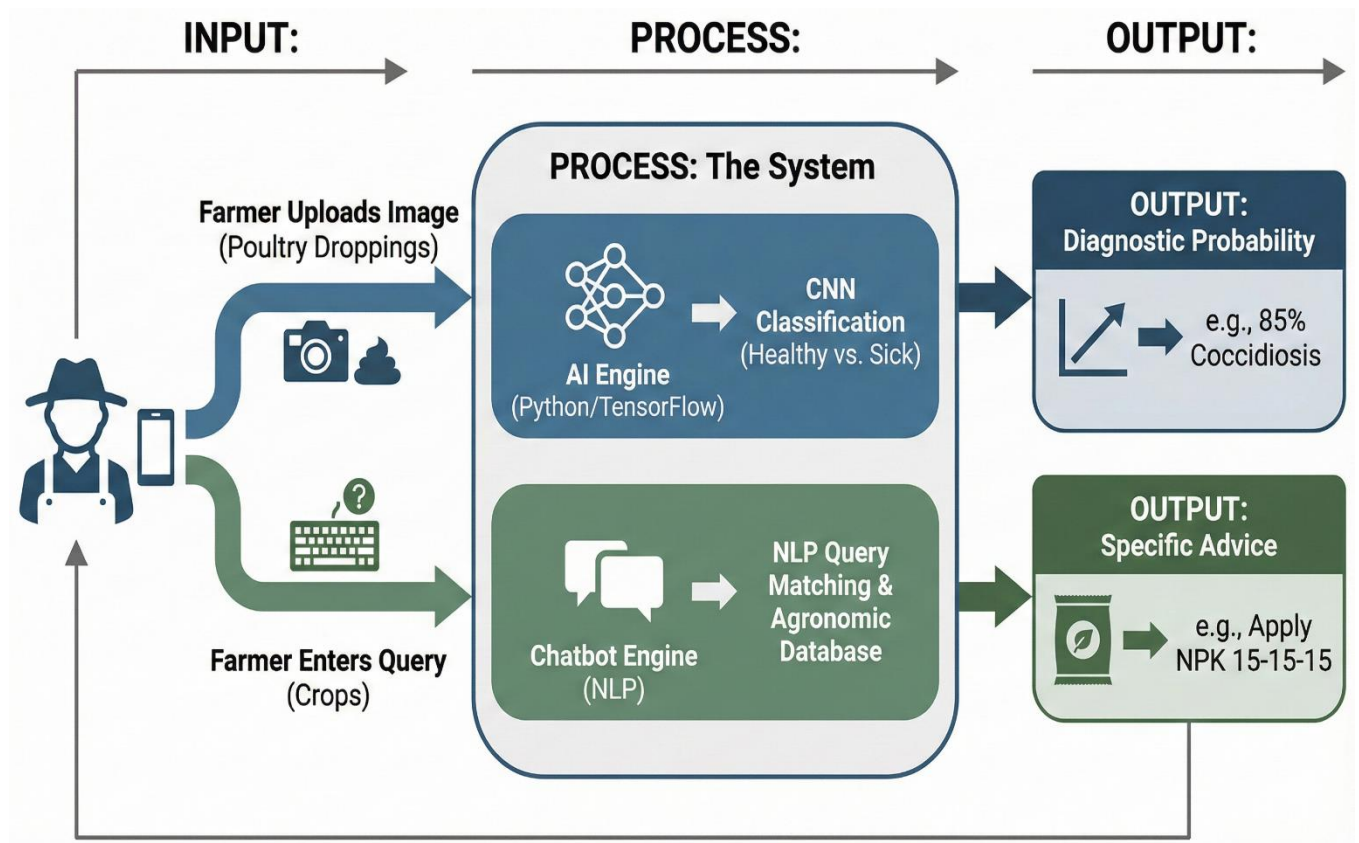
- **Fecal Image Analysis:** Degu and Simegn (2023) demonstrated that analyzing chicken fecal images is a non-invasive and highly accurate method for detecting internal parasites. Their work provides a validation for the core mechanic of this project.
- **Lightweight Models:** A critical challenge in rural deployment is hardware limitation. Islam et al. (2025) proposed a "lightweight" model approach using multi-color space feature optimization, achieving high accuracy (95.85%) without requiring heavy GPU support, which guides our technical choice for the web deployment.

**7.3 AI in Crop and Soil Management** Babatunde et al. (2025) describe AI as a "game-changing technology" for sustainable crop management. Furthermore, surveys of deep learning applications in agriculture indicate that algorithms like Random Forest and SVM have been successfully used for crop recommendations with high accuracy (Book Chapter Survey). Khaliq et al. (2025) further support this by proposing integrated approaches where AI analyzes soil and crop data to recommend specific fertilizers, a feature mirrored in our proposed chatbot.

### Conceptual Framework

The system operates on a standard Input-Process-Output model:

- **Input:** Farmer uploads an image of poultry droppings (for the AI) or enters a query about crops (for the Chatbot).
- **Process:**
  - The **AI Engine** (Python/TensorFlow) processes the image using a CNN to classify it (Healthy vs. Sick).
  - The **Chatbot Engine** processes text using NLP to match queries with the agronomic database.
- **Output:** The system returns a diagnostic probability (e.g., "85% Coccidiosis") or specific advice (e.g., "Apply NPK 15-15-15").



## Methodology

### 9.1 Research Design

This project follows the **Agile Software Development** methodology, allowing for iterative development of the marketplace, AI model, and chatbot features.

### 9.2 System Architecture

- **Frontend:** Built with **React** for a responsive user interface.
- **Backend:** A **Node.js** server handles the application logic, while a **Python (Flask/FastAPI)** microservice hosts the AI model.
- **Database:** **MySQL** is used for storing user data, products, and chat logs.

### 9.3 AI Model Development

- **Data Source:** We will utilize public datasets of poultry fecal images, such as the Zenodo dataset referenced by Degu and Simegn (2023).
- **Algorithm:** We will implement a Convolutional Neural Network (CNN). Following the recommendations of Islam et al. (2025), we will prioritize a lightweight architecture (like

- MobileNet or a custom shallow CNN) to ensure the system remains fast and responsive on the web.
- **Training:** The model will be trained to classify images into "Healthy" and "Coccidiosis" classes.

## 9.4 Chatbot Development

The chatbot will be developed using a retrieval-based approach. It will draw on a knowledge base constructed from best practices identified in the literature, such as the crop-fertilizer recommendations discussed by Khaliq et al. (2025).

### Expected Outcomes

1. A fully functional web application integrating E-commerce, AI diagnostics, and Advisory services.
2. An AI model capable of detecting Coccidiosis from fecal images with an accuracy target >85%.
3. A reduction in the "information asymmetry" faced by farmers, as described by Addison et al. (2024).

### Work Plan (Timeline)

| Phase                          | Activity                                                   | Duration    | Deliverable            |
|--------------------------------|------------------------------------------------------------|-------------|------------------------|
| <b>Phase 1: Research</b>       | Literature review, data collection (poultry feces images). | Weeks 1–3   | SRS Document & Dataset |
| <b>Phase 2: Design</b>         | UI/UX Prototyping and System Architecture design.          | Weeks 4–5   | Wireframes & DB Schema |
| <b>Phase 3: Development I</b>  | Core Marketplace development and Chatbot Implementation.   | Weeks 6–9   | Functional Marketplace |
| <b>Phase 4: Development II</b> | ML Model Training (Coccidiosis CNN & Feed Optimizer).      | Weeks 10–15 | Trained ML Models      |
| <b>Phase 5: Integration</b>    | Connecting ML models to the Web App.                       | Weeks 16–18 | Beta Version           |

|                                          |                                                           |             |                               |
|------------------------------------------|-----------------------------------------------------------|-------------|-------------------------------|
| <b>Phase 6: Testing &amp; Deployment</b> | Bug fixing, Field testing with farmers, and Final Thesis. | Weeks 19–21 | Final Project & Documentation |
|------------------------------------------|-----------------------------------------------------------|-------------|-------------------------------|

## Conclusion

This project proposes the design and implementation of an integrated web-based agricultural ecosystem to address the fragmented nature of current digital tools for Ghanaian smallholder farmers. The study intends to solve the "triple threat" of limited market access, scarce veterinary support for poultry, and insufficient agronomic advice. The general aim is to create a unified platform that facilitates e-commerce while providing AI-driven diagnostic and advisory support. Specific objectives include developing an e-commerce marketplace, implementing a CNN model for Coccidiosis detection, and creating an interactive chatbot for real-time guidance.

Following an **Agile Software Development methodology**, the project will utilize a modern tech stack (React, Node.js, Python, and MySQL) to build the prototype. This study demonstrates practical relevance by bridging the digital divide and empowering farmers to produce efficiently and sell profitably. By implementing lightweight AI models, the project ensures feasibility in low-resource environments, making it a valuable and achievable contribution within the scope of a BSc undergraduate study.

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